

FINITE PRODUCT

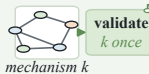
Flat: repeat every cell

	$p1$	$p2$	...	pP
$k1$				
$k2$				
kK				

K mechanisms $\times$ P views
 $\times$ A deterministic agents

p can change o
 memory + action
 may change
state-only key unsafe

1 VALIDATE ONCE

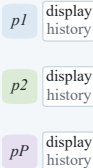


Presentation contract
 p leaves physics fixed
 terminal rules fixed

reuse level 1
 one graph build per k

2 EXACT STEP REUSE

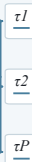
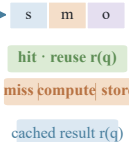
presentation-local display



s world $\cdot$ m pre-memory $\cdot$ o observation
 display history + traces remain local

private trace

exact step index
 fixed (k, a) ; key q



EXACT OUTPUTS

$\tau(k, p, a)$
 declared traces

$\sigma(k, p)$
 six pair bits

$\rho(k)$
 all- p AND

Example:
 all-view query
 exact cover

INDEPENDENT
VALIDATION

separate interpreter
 rebuilds full semantics

compare every valid flag
 trace + six-bit σ
 Flat vs LayerProbe

traces + signatures

=

declared outputs
 agree exactly
 full-domain agreement